Magnus Rönn

Architectural quality in competitions

A dialogue based assessment of design proposals

Abstract

This article is the outcome of a comprehensive study of architectural competitions in the Nordic countries (1999-2000). The vital content of the research is derived from a series of meticulously conducted interviews with key actors including 18 highly qualified and experienced experts from the jury members representing clients, architectural associations and competitors. The study refers to quality as a key-concept and a main source of conceiving, judging and selecting a prize winner. It is possible to articulate how architectural quality issues are met by jury members; how they are being communicated; and how a winner is nominated. The study provides insight into how the concept of quality in architectural design is understood in practice. Quality is identified through design criteria in a dialogue-based assessment of architecture and urban design projects. The assumption is that the judgment and evaluation of entries in competitions are strongly connected to the leading values, norms, regulations, organizations and traditions in Scandinavia. When quality is contextually bound, the assessment becomes a question of how the solutions fit the specific plot. These issues in a competition process cause uncertainties and discrepancies in judging and selection. However, while the main role of jury members is to agree upon the most appropriate solutions, they finally succeed in designating the best entry through their cumulated tacit knowledge and well-trained eyes. Competence and consensus are therefore two essential factors that make jury members feel confident in their final choice of a winner.

Keywords: architectural competition, quality assessment, architectural quality, design criteria

Introduction

Scandinavia enjoys a tradition, which is more than a hundred years old, of arranging architectural competitions for the selection of the best design practices. Every year, about a hundred architectural and urban design competitions take place in the Nordic countries. The competition system is a recognized instrument in Denmark, Norway, Finland and Sweden used to highlight quality in architecture and urban design. It gives opportunities for talented architects to participate and express their visions, to demonstrate their professional skills, and to be rewarded, admired, and endorsed publicly. The present article is the conclusive part of a research project at the School of Architecture aimed at developing a theoretical discourse on quality issues in general, and on design competitions as a system for evaluating architectural quality, in particular.

The article consists of three somewhat parallel themes. Regarding the first theme, the article deals with *architectural quality* with reference to the concept as it is conceived, defined, and used by experienced architects in their practice. It is based on the analyses of the language and terminology experts use when they try to express themselves about quality in architecture and urban design. This theme is developed by obtaining information from a series of face-to-face interviews. Direct quotations are presented when referring to particular queries in the questionnaire about the definition of quality and how jury members assess it. The concept of quality, as defined in the field, is then analysed and discussed with reference to the answers from the informants. The most essential issues are: What is architectural quality? How is quality defined, understood, translated, and evaluated through a competition process?

In relation to the second theme, the competition is analysed as an instrument for *judging quality* and using design criteria in architecture to define differences in design proposals. Architectural competition is seen by the profession as an old, effective, and reliable instru-

ment for establishing dialogues among jury members representing professional experts, clients, and even end-users, during the selection and assessment of architectural design proposals. Architectural competitions are usually organized in order to search for the most appropriate solutions to a given task. How do experts deal with the selection of a winner from among many different entries? What are the principal criteria for assessment, evaluation and selection of winning entries?

The third parallel theme in the article deals with the *judging process* in competitions and how the jury's work is organized. Six specific steps are identified as being crucial to the outcome. Important questions related to this theme are: What determines how the steps are categorized? How is the jury's work organized and how is the assessment process carried out in competitions? My intention is to show how the jury singles out a winning proposal. A basic part of the study has involved investigating the way practitioners talk about quality, how jury members point out good solutions to design problems, and how architects demonstrate quality during the selection process of a competition. There are numerous professional perspectives, cumulated experiences, profound types of knowledge, and a repertoire of complex concepts on architecture and quality that need to be clarified. Language, design, and judgment are three interwoven elements that are very dependent upon each other. They indicate a professional culture. It is my task in this article to illustrate how architectural qualities are presented, debated, articulated, reasoned, and assessed inside the jury room.

Methodology

In 2004, the School of Architecture at the Royal Institute of Technology, Stockholm, received funding for the research programme called *Quality assessment and quality improvement*. Within this framework two studies of contemporary competitions were undertaken (Kazemian, Rönn, & Svensson; 2005; 2007). Magnus Rönn was project leader for both studies. First there was a review of competitions held in the Nordic countries (Denmark, Norway and Sweden) with a special mapping of competitions in Finland (see Figure 1). Second, Charlotte Svensson has in an ongoing PhD project investigated the assessment process inside the jury room, focusing on how the jury members selected a winner (see Figure 2). These two projects will be referred to as the *Nordic Study* in this article. I will present some findings from the research program. The purpose is to demonstrate how we can understand quality in architecture, the evaluation of design proposals, and the assessment process in competitions.

Figure 1. An investigation of contemporary architectural competitions in Finland.

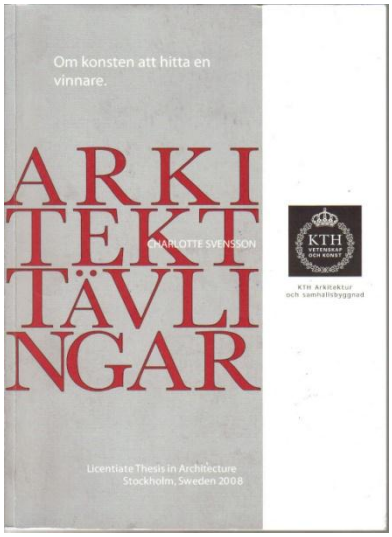

Figure 2. A licentiate thesis on competitions by Charlotte Svensson.

The methods for this investigation include 'close reading' of documents (architectural policy programmes, competition documents, and transcripts of interviews); conceptual analysis of how quality is expressed by practising architects; and a follow up of competitions held in 1999-2000. During these two years, 199 competitions were identified in the Nordic countries. This survey concerns competitions approved by the architects' associations. The competing architect firms and the organizers were consulted by letter for control statistics and the home-pages of the architects' associations were used as well. 'Close reading' in this context means that I have highlighted key words in the investigated documents and compiled them into meaningful categories. Transcripts of interviews have been examined and compared with statements in briefs and jury reports.

The starting point in the Nordic Study was the assumption that practice reveals how the concept of quality is used in architecture and urban design. The ways in which practitioners think and speak about quality in jury reports, as well as their approach to reviewing and judging them have a fundamental effect on how the concept is conceived, communicated, and understood. For this reason, 18 experienced jury members were interviewed along with architects and urban planners in Finland, Norway, Denmark, and Sweden. The informants were selected based on their personal experiences with architectural competitions. They are among the well-known professionals with many years of active involvement as members of competition juries selecting winners. They represent the three groups of key players in the competition system:

1. The organizers; promoters, and clients
2. The competing architects; firms and project groups
3. Architect associations; competition administrators and inspectors

The interviewees were chosen for their professional experience. They have been involved in over a hundred competitions as competing architects, jurors, and representatives from organizing bodies. A questionnaire on competitions, with questions covering aspects from the programme to the nomination of the winner and drawing up of the statement by the jury, was developed for the interviews. The interviews lasted from one to two hours. The transcripts of the interviews were printed out and verified by the interviewees. Together with competition documents (briefs, proposals, meeting notes, and jury reports), these interviews give a good picture of the competition system in practice. The construction sector in Finland and Denmark

competed slightly more often than in Sweden and Norway. According to the follow up, Finland and Denmark seem to have a stronger competition culture than the rest of Scandinavia.

Competition and quality issues

I understand architectural quality to be an open, valuing and norm building key concept, which – through several meetings, conversations, debates, suggestions and legal provisions – will be specified and interpreted. Exhibitions and publications are among the major communication channels used to reach the outside world. Coherence, context, tradition, and consistency affect our understanding of quality. How we conceive architectural quality has an important impact on both the design of a project and the assurance of its quality. It is also important for the elevation of the overall quality standards in society.

For professional quality assurance, three moments are very decisive. One defines quality; the other brings about quality; and the third supervises and controls it. Experience of quality is often turned into demands for it, so as to promote it in the early design stages, with the hope of minimizing the eventual mistakes and shortcomings of a design solution. The underlying idea is that the quality of an artefact – an architectural product – can be predicted and foreseen as early as possible through presenting plans and models (Rönn, 2007). Architectural quality is, however, more than just error-free and inflexible products. A zero error design is not the same as a good, high quality, well-designed, product. Quality should be seen in context, as a positive increment in the environment.

There is no definite and ultimate answer to the fundamental question of architectural quality. Qualities in new forms, types, and contents will constantly appear in architectural design. The quality concept will take on new meaning in time and in different places, even for those who believe that values are eternal. The concept is dynamic and complex and the dualism becomes clear when architects attempt to give their personal answers when asked what distinguishes good architectural quality. This approach to understanding quality in architecture and urban design can be found in the national policy programme which developed in Europe during the 1990s starting in the Netherlands (1991), Norway (1992), and Denmark (1994)[1]. These documents can be found on the homepage for the European Forum for Architectural Policies (2011) (See Figure 3).

Figure 3. The logo for the website on architecture policy in Europe. The programs have an aesthetically oriented concept to promote quality in architecture and urban design.

All professions have their own specific ideas about quality and these are promoted within their particular areas of activity. Their knowledge of quality is influenced by the kind of education, training, and expertise they acquire and by the school of thought or the working culture to which they belong. This is why the informants in the Nordic study represent a common culture.

Architectural education aims at producing architects with sufficient skills to suggest and create good characteristics in architectural design. Through years of practice, their specific judgment and ability can help to develop, visualize, evaluate, and identify appropriate de-

sign solutions that reflect quality. Their long education, training, and experience give architects sensitive eyes to observe, draw, show, and finally compare and judge what good design quality is about. To design quality and to assess it, constitute two dependent sides of their professional competence.

When it is applied to professional performance, the concept of quality assumes a collective character and content. Quality issues as professional practice are linked to giving advice. We rely on quality judgments that are expressed by well-educated and competent professionals. Laymen with some degree of confidence about their personal experiences can also comment on quality aspects. But professional knowledge appears to be vital when the aim is to create quality in a very complex design task and context.

In order to maintain professional reliability we should know what distinguishes a good design result. We need to point out a model for a solution with confidence and should be able to indicate the result, as well to attain it. This is not going to be an easy task. In this way, quality becomes a matter of applying appropriate methods, regulations, and knowledge that are reflective, either through tacit knowledge or articulately. Professional practice involves following the rules of how to achieve quality. But what is quality? How can we understand the concept of quality in architecture and urban design?

Quality in Design programs and regulations

The Swedish Cultural Report SOU 1995:84 pointed out that architecture and design are cultural expressions that are vital to people's well-being (Kulturdepartementet, 1995). The report suggested therefore, that the government take the initiative to formulate an architectural policy. A new political era was thereby created. Two years later, in 1997, *Forms for the future* – the Swedish Action Programme for Architecture and Design – was presented (Kulturdepartementet, 1997) (see Figure 4).

Figure 4. The front page for the Swedish Architectural policy.

The National Policy contains a special part entitled 'What is Quality?' that aims to reframe and redefine the concept. According to *Forms for the future,* quality is seen as something positive and transformational. 'Good quality' in architecture and urban design, according to this policy, is a concept with the following characteristics:

> Good quality cannot be defined once and for all. Our experiences, knowledge and values influence how we look at quality. Quality is also dependent upon situation and time ... This

> means that we reconsider what is good quality according to how our values and needs change. Good quality in our designed environment requires a constant critique of what we need, what shall be prioritized and what solutions are best suited to meet these needs. A building or an object is designed in order to be used ... But it is not just a matter of solving practical problems. In good architecture and design there is always a basic idea of an aesthetic nature that has to clarify the object's properties ... In architecture and design there is a combination of different requirements and needs ... in order to get a well-functioning wholeness: function, usefulness, housekeeping, with natural resources, good design, appropriate technology, and cost efficiency... Buildings and objects shall enrich our lives today and they shall respond to different people's different needs. They must also cope with changes over time and can be adapted and also used in changing situations. What we produce today is a footprint of our time and something we shall be able to proudly pass on to future generations. (Kulturdepartementet, 1995 pp. 12-13)

The global target for every quality-enhancing activity in *Forms for the future* is to create appealing solutions to design problems. 'Good quality' is what is looked for, a value-laden concept from a cultural perspective. Quality has an aesthetic dimension, which can be pointed out, shaped, assessed, and experienced. The policy considers quality in architecture and urban design to be a concept that can be mediated by models, illustrative examples, interesting cases, and critique. Quality, interpreted from such an emotional perspective, differs from the measurable 'right quality', a technically oriented definition of quality that is common in industrial construction and the building sector.

However, both definitions of quality can be found in the Swedish Planning and Building Law (Socialdepartementet, 1987), where architecture and urban planning stand more for the emotional understanding, while engineering and industrial construction consider the technical aspects of quality. The different ways in which the concept is perceived might be a main source of misunderstanding between architects and engineers. There is disagreement in the building sector as to what architectural quality *is*, how appealing environments can be *created,* and how they should be *judged*.

In 1999, the Swedish Planning and Building Law was revised with a supplementary section on aesthetic issues in planning and building (Socialdepartementet, 1987 Chapter 3, §§ 1 and 9). The new paragraph was a direct result of the Proposition of the Acting Programme for Architecture and Design of 1997 (Kulturdepartementet, 1997). According to the new supplements, building design should be 'aesthetically appealing,' and should have specific 'characteristics,' and 'artistic values' that should be carefully considered and revitalized during the reconstruction. Planning should be carried out with respect to 'aesthetic values.' Quality in architecture and urban design becomes a matter of judgment that can be best experienced and appreciated in a specific context. Architectural quality is not a general matter but is connected to a particular project which fits a specific plot.

The technically-oriented quality concept in the Swedish Planning and Building Law (Socialdepartementet, 1987 1994:852 Chapter 9: § 9) emerged in 1994. Through a supplementary paragraph, the developers became the party designated to be responsible for building quality when legislation called for certification and control to ensure that the 'right quality' was produced. The idea behind the revised Law was not only to create new environments with good architectural quality, but also to minimize mistakes during the building process. Responsibility for quality is included in a specific engineering tradition that regards quality as encompassing characteristics, functions, and implementation that can be measured, assured, practised, and controlled (Nashed, 2005; Nelson, 2006).

In both concepts quality is considered as something desirable. In the early design phase, there are no guarantees that mistakes that affect the end users' quality experience can be avoided. It is also impossible to approach good architecture just by exactly measuring and calculating the technical details of a building. It is, of course, fine to have zero mistakes as a target for good building quality, even if drawings that are free from mistakes cannot guarantee

architectural quality when plans are implemented and experienced in a built environment. Good quality must be something more than zero mistakes in architectural drawings, design and building construction. Architectural quality assumes that the proposed plan and building are to be regarded as a positive addition to the environment (Rönn, 2007).

Architectural quality perceptions

In order to find out how experienced practitioners examined quality in competitions, the Nordic Study included interviews. One special section of the questionnaire was about quality issues and the ways in which architects perceive *architectural quality*. The informants were asked to define their own understanding of the concept. The result consisted of illustrative answers showing their typically professional views on the concepts. Architectural quality was a professional, cultural, and historically defined key concept. The following quotes give a picture of how the concept of architectural quality is understood by architectural practitioners and can be divided into five categories.

First, the concept of quality, through a legacy from Vitruvius, seems to be an indivisible unity of form (*venustas*), function (*utilitas*), and construction (*firmitas*). There is a 2000-year-old tradition that appears to be very much alive and powerful, a canon to generation after generation, that architects ominously return to in their tasks and missions. Good architecture is a combination of form, function, and construction. Quality appears as wholeness in the way this global unity is created.

> I keep to Vitruvius, who says that architectural quality is about aesthetics, function and construction. Everyone can draw a house that functions and stands for a period of time. But a house needs to be pleasant and beautiful as well. Architectural quality comes up when aesthetic, functional and constructional problems are optimally solved (J. Christiansen, personal communication 2005).

Second, architectural quality is perceived in relation to its context and its special preconditions. Consistency and coherence are decisive for judgment. The public space, infrastructure, buildings, and the surrounding landscape shape the grounds for judgment of quality. The place has a spirit that has to be discovered. Architectural quality is targeted to fit the context, the plot, and the surroundings.

> For me, architectural quality is about how to place a house on a plot, how a plot is used, how to create room and space, how to move in a building, how to see in and to see out ... A building with quality stimulates me; makes me curious, happy and joyful (B. Holm, personal communication 2005).

> Quality … should be something that has its own life and sticks out in a way. Architectural quality often deals with interplay between surroundings and context and can be observed and experienced in a positive sense (T. Nordberg, personal communication 2005).

Third, the concept is linked to a mystical aesthetic feature in architecture, which is difficult to explain. Quality is a positive and surprising experience and expresses a personal feeling of approval that can be found in artistically solved design problems. Architects don't identify winning entries in competitions by research or reading peer-reviewed articles, but by an experience of quality, which is felt as a kind of wholeness. It is the feeling combined with skilled eyes and professional competence that identify quality.

> Quality, at a fundamental level, is about norms and demands that need to be fulfilled. But there is a peculiar subjective side on a higher level that deals with judgment, aesthetics and personal insight … Quality is something more than demands that should be fulfilled … the

> concept of architectural quality carries some mysticism in itself (M. Mäkinen, personal communication 2006).

> Architectural quality has always been about making a surprise. It means a new combination of traditional materials put together in a way, which gives a feeling of satisfaction to the one who is experiencing that space. It is dealing with interpretation of factors that might not have been known before (P. Huotelin, personal communication 2006).

Fourth, architectural quality is a matter of expressing timeless values in a way that is typical for its contemporaries. Originality and novelty attain their right values accordingly. There should be an ideal model in architecture that can survive the waves of shifting vogues. Quality is seen as the result of selection and acknowledgement. Approval from colleagues is seen as a confident sign of assured quality. There is an additional reason why architectural history is perceived as being alive, useable, and important in providing knowledge to architects for professional endeavours. In this regard, performance of quality can put jurors in a difficult situation. The jury has to identify the quality of an architectural design proposal at a very early stage; this can be affected and changed later on when it is built, when it is in use, and when it is compared to other buildings or environments.

> Architectural quality is a concept with a double meaning. On the one hand, architectural quality is a professional practice. On the other, quality is a matter of subjective judgment that is the outcome of the best competence that gradually gains its right value … Looking through the historical light, there is often a consensus on quality in architecture. From this point of view, architectural quality exists in buildings that over time, could survive varying judgments. The subjective side of architectural quality is an expression of approval. Architectural quality has appropriate solutions that are appreciated by capable, professional architects (M. Sundman and P. Pakkala, personal communication 2006).

Fifth, architectural quality has a usability value. Quality is ascribed to buildings or environments that possess appropriate forms and fulfil required technical specifications. Good architecture often appears with multi-aspect purposes and uses. Aesthetics and techniques are combined and coordinated for a practical solution. Quality becomes a practical question of material, construction, sustainability, and usability as well as a test bed for how design corresponds to the spatial needs of particular activities and the users concerned.

> Quality is a practical question. It is difficult to separate different qualities from each other, but still we can talk about architectural quality, technical quality, or functional quality. For instance, quality can be seen as practical when it can be separated from architectural or functional. As architects we should be able to distinguish and include all these attributes and be more responsible for architectural quality as a whole (T. Kurkela and J. Murole, personal communication 2006).

> Quality means that the project must operate. The functional, structural, and technological solutions in the project must be in order ... It is important that the project be sustainable over time; the society changes as well as the architectural design, but the technical solution should be reasonably stable and work in the future (K. Beite, personal communication 2005).

The professional language among architects is clear in Scandinavia on one very crucial point. In four out of five categories, quality is an aesthetically oriented concept. It is typical of how architectural quality is understood in competitions. This fact can be explained both by the competition being about the design in the beginning of the production process, where aesthetic dimensions are in focus, and by the position architects have in the building sector. Practising architects deal with quality issues in the early stages of architecture and urban design projects, when finding a primary generator, a fundamental idea, is vital for the solution of design

problems (Drake, 1979). Only when architectural quality is linked to the final purpose, use, and practical solution, does the concept take on a technical theme for the profession. By then the award winning proposals have already been chosen.

Design criteria

The architects interviewed in the Nordic Study were also asked what criteria they used as a basis for their judgments of quality in competitions. Analyses of collected data pointed out two different types of design criteria: *evaluation criteria specific* to the project and *general criteria*, used in different competitions. Evaluation criteria linked to a specific project are based on the written, distinguishing features described in the competition programme. These design criteria vary from competition to competition. But there is also a stable pattern – a number of criteria which appear time and again in competitions and which influence the jury's judgment of quality on a deeper plane. I call these general criteria since they have a broader area of application and express a deeper understanding of how quality is identified.

In principle, all competition entries will be judged by general criteria even if these are not specifically outlined in the competition programme. General criteria are commonly acknowledged signs of architectural quality, rooted in professional experience. They are an expression of tacit knowledge[2] among architects. The general criteria were identified when transcripts of interview data were compared with 'close reading' of jury statements, selected for further investigation in the research project.

The analyses showed that there were six general design criteria, which reappear when judging competition entries. These criteria reflect an understanding of how the jury members should proceed to determine the decisive differences between the competition entries:

1. *Wholeness and fundamental idea:* How has the competitor solved the competition goal on the whole? Is there a powerful design idea? To what extent has a strong fundamental idea and an appealing design been combined with functional demands, durability and economy.

2. *Coherence and surroundings:* How well does the proposal fit the site? Is the scale appropriate? How does the design blend in with the neighbouring buildings and the surrounding landscape?

3. *Entrance position:* How has the competitor solved the entry into the area, site, and buildings? What is the relationship between the outside traffic and the inner movement pattern in the area and building?

4. *Suitability and functional set up:* How has the competitor solved the spatial organization? How does the proposal work in regard to the end users' planned activities? How have the end user's functional requirements been met?

5. *Economical and technical solutions:* How is the contribution technically produced? Are the system solutions, constructions and materials safe, buildable, and economical?

6. *Development possibilities:* To what extent can the proposal be further developed? Can some of the shortcomings be corrected and other solutions improved without losing the fundamental idea and without compromising the architectural quality?

The design criteria are part of an assessment based on dialogue and they have two principal functions. They tell the jury members *what* is important to judge and *how* to proceed. The first step is to direct the jurors' attention: This is the 'what'. The second step is a question and represents the 'how'. Since there is no single, clear-cut solution to the architectural quality

question, an open attitude is necessary. The jury acquires knowledge by posing questions about the proposal. The questions reflect the inquiring nature of the criteria. They are, in turn, the consequence of the multifaceted nature of the competition. The jurors are confronted by several interesting solutions to the design problem in architecture, urban design, and town planning; this becomes very obvious in an open competition with many entries. It is not possible to arrive at a sole, 'objective', best solution. Nor is the outcome of an assessment in competitions a result of luck as Kreiner (2009) proposes. Not at least from the jury's perspective.

I believe the answer to architecture's quality questions lies in a well-balanced entity of dilemmas (Rönn, 2010). The evidence for this statement can be found inside the jury room (Svensson, 2009). The winner is chosen through architectural critique by jurors striving for quality in the proposals. The architects on the jury act as guides for the laymen. Their minority position in the jury does not reduce their professional power. It can instead increase their position as experts. It is their role as undisputed experts on design that, in this context, gives architects a powerful position among the laymen of the jury. The majority of the representatives for the client in the jury are experts in fields other than architecture.

Discernment, experience, and a trained eye contribute to judging competence. The architectural interpretation of the proposal will be even richer and further enhanced when it is examined from several angles. That is the reason the jury is made up of members from various fields: architects, clients, promoters, and users.

Competitions as quality judgment process

Architectural competitions are examples of judgment where a group of competent people are designated as jury members with the task of finding the best design solution among many entries. The design proposal is presented to the jury with the designer remaining anonymous. The selection process takes place in several stages. The jury discusses and gradually comes to its final selection by considering each entry and its potential for fulfilling the expectations outlined in the competition programme. Selection is carried out in the form of negotiations based on the quality of the entries. A proposal will remain in the competition as long as it is defended by jury members.

The best entries will be designated and sorted out through discussions. Quality assessment will be based on the design criteria in the brief that reflect the demands and goals of the design task. Interpretations of the programme by professionals based on tacit knowledge also appear and are important for ranking in the final stages. In the end, a handful of selected entries, which the jury collectively finds appropriate, will remain. The process is a search for future quality. One single entry is usually pointed out as best by the majority; in the end there is consensus. It is very seldom in Scandinavia that the jury members disagree upon the winner in the final report. In only two out of sixty-six competitions in Finland during 1999-2000, did the jury decide to split the first prize (Kazemian, Rönn & Svensson, 2007).

A majority of the jury members represent the organizing body. The organizer selects a chairperson, a secretary, a competition functionary, and different representatives for the client. At least a third of the jury members must have the same competence as the competitors (European Parliament and of the Council, 2004). Normally, architects dominate the jury's work even if they are in a minority. In Scandinavia, the professional associations select two architect judges to be on the jury.[3] They represent the professional interest in the competition and present the proposals to the rest of the jury.

The competition functionaries have to maintain contact between the jury and the competitors. Any complications and complaints in a competition process will be dealt with by these functionaries. Functionaries, like other members in a jury, have to maintain professional secrecy throughout. Because of anonymity in the competitions there can be no direct communication between the jury and the competing project groups.

The process differs slightly depending on the competition's form. Usually it takes four to eleven months to process the activities in a competition. This period covers the time for programme preparation, additional development of entries, and the judging process. There are usually three to six competitors in a competition based on invitations. The open competitions normally vary from thirty to three hundred participants in Scandinavia, as it did in Finland during 1999-2000. In open competitions five to six official meetings are needed to nominate a prize-winner. In invited competitions, four to five meetings are normal. Between meetings, working groups meet to study the entries in detail and prepare for the next official meeting.

The jury members have the task of identifying, describing, and ranking the competition entries, taking into account their quality. Two principles seem to dominate the assessment: Poor proposals are rejected; they get no support. Good solutions are emphasized; they become favourites and will therefore stay in the competition. The assessment process is driven forward as the jury members point out their favourites. In open competitions it is usually easy to arrive at a handful of especially interesting solutions to the competition task. This is the general experience expressed by informants in the Nordic Study. According to the architects, good solutions in design easily become apparent to the jury. The skilled eye is to be trusted.

The difficulties first arise at the final judging when the jury has pointed out a best overall entry. One single design solution is seldom superior on all counts. If it is difficult to come to an agreement, the jury will discuss the matter once more with the aim of reaching a consensus. This is clear advice on the part of the informants. If the jury cannot come to an agreement there is a risk that the competition task will remain unsolved. One way to work out the difficulties at the end is to make scale models and highlight the competing proposals on the competition's site. Then the jury will see which of the design solutions fits the plot best. The eye becomes the final decision maker – not the best argument.

The jury's judgment of the competition entries has its point of departure in the demands set out in the competition brief. The assessment follows a specific pattern. It is a general pattern for evaluating quality in the Nordic Study. The informants describe the judging as a process involving six stages (Kazemian, Rönn & Svensson, 2007):

1. *Submissions check:* The jury begins the evaluation process as soon as the entries arrive. This is a formal control that is regulated in the competition's procedural rules.[4] In order to be accepted for a competition, all entries should be checked in on time and fulfil all requirements prescribed in the competition programme. In open competitions, many entries will be disqualified during the initial control stage.

2. *Order of work and examination:* The jury decides how its work is to be carried out and members begin to acquaint themselves with the proposals. The jury members walk around in the gallery, individually or in groups, to study the displayed entries. Usually the jury forms smaller groups in order to scrutinize the proposals and prepare for coming meetings. The nature of the competition task determines the need for working groups and specialist support. It is primarily for competitions dealing with urban planning and special purpose buildings that the jury enlists specialists. Cost calculations are used in the final assessment to see if there are any decisive economic differences between the competition proposals.

3. *Choice and preliminary assessment:* The architect members of the jury make an initial quality assessment of the proposals, often in consultation with the competition secretary. This is a professional evaluation that results in a selection of proposals that are considered to have appropriate solutions for the competition task. Selected proposals are presented to the jury as a whole. This assessment is preliminary and nothing prevents changes from being made later. The jury members can reconsider proposals

that have been sorted out. The final choice is made in stages, as part of a searching process that is initially open in character.

4. *Presentation of interesting contributions:* The architect members of the jury formulate an unbiased and professional description of the selected competition contributions. If necessary, this list is complemented after discussion among the jury. The members emphasize their favourites. Then there is an evaluation of the proposals that have been singled out. The architect members have a special expert responsibility with regard to emphasizing good solutions and pointing out shortcomings. This is a matter of identifying the qualities behind graphically seductive presentations of environments and photomontages full of people. A compact cluster of proposals now begins to crystallize. The jury members then go back to their working groups and continue to scrutinize the selected competition proposals.

5. *Ranking:* The next time the jury meets, as a basis for their discussion, there is usually a suggestion as to the ranking of the competition proposals. The need to find a winner forces the members to make value judgments. Doubt must also be overcome. A critical situation occurs when all jury members are forced to present their personal opinions. It can be especially difficult to distinguish between the best proposals. The jury shall take into account how the proposals can be further developed and make a future-oriented overall assessment where shortcomings of a detailed nature can be ignored. Sometimes before the jury finally decides on a winner, specialists are called in to provide complementary information. In open competitions the jury awards 1st prize, 2nd prize, 3rd prize as well as some other awards for well-designed proposals. The prize sum will then be shared among the prize-winners. Regarding competitions by invitation, it is sufficient to proclaim a single winner, as all participants receive the same fee.

6. *Final decision and jury report:* The jury completes its quality assessment by issuing a recommendation to the organizer and providing an account of the winning proposal, other prize-winners and overall critique of the competition. The jury's verdict contains two types of critique about the competition. This is partly a general critique of the competition and partly a critique of the individual prize-winning projects. Sometimes the jury members point out so many shortcomings in the winning proposals that their choice seems surprising. However, this criticism is interpreted as advice prior to an impending detailed design commission. It is the forward-looking function of the criticism that causes the jury to point out any uncertainties and shortcomings in the winning proposal that need to be reworked in the implementation phases.

This description is a model indicating how the assessment process in competitions is organized, but also how the assessment should be organized in such a way that the jury can find a winner among the entries – a design proposal that can be considered to offer the best overall solution for the competition task. According to informants, there is a clear correlation between how the assessment process is executed in specific steps and the jury's ability to identify a first-prize winner. The assessment starts with a formal control process and concludes with the jury formulating a declaration containing a decision and architectural critique of the competition and awarded entries.

But how does the jury 'know' that the right contribution has won? On what grounds can a jury feel sure that it is 'the best solution' that has been rewarded? The jury works through its task successively. This is a means of dealing with the uncertainty in the assessment process. The jurors are cautioned to be careful about forming all too certain judgments

at an early stage. Identifying the winner is the result of a long process of negotiation. What is it that enables the jury to make a final decision and feel sure about its choice?

The architects interviewed argued that it was vital that the 'right' proposal win the competition. This choice is scrutinized by colleagues, provides employment for the winner of the first prize, and shapes the future environment. The informants meant that it is competence and consensus that make the jury feel certain that the best solution for the competition task has been nominated for the first prize. It is not more difficult than that. The architect judges have to be skilled members of the association with good reputations. The organizer must rely on their expertise. The 'right' design will then be the one the jury agrees upon. Consensus gives security. This is an indication that the jury has completed its task satisfactorily. Reservations, on the other hand, are an indication of doubt. A shared second prize instead of one winner may lead to the organizer having doubts about the result of the competition, even if the reason for this can be traced back to ambiguities in the competition brief. If the jury cannot point out a winner the problem falls back on the organizer.

At the end of the competition, the jury formulates a declaration – a recommendation to the organizer. This declaration should contain two kinds of criticism: general criticism of the competition as a whole, and individual criticism of the prize-winning solutions. It is the advisory and forward-looking function of the criticism that induces the jury to point out shortcomings and unclear points in the winning proposal that need to be adjusted. The aim is to contribute to a good end product. An informant in the Nordic Study also suggests that critique is crucial in the assessment process in order to find the best solution (Kazemian, Rönn & Svensson, 2007). In the interview, Mäkinen says:

> To me, questions of architectural assessment have always been important. The jury members – particularly the professional architects – should have an understanding and experience of architectural critique. Someone who makes a professional statement about architecture is responsible for their judgment. Architectural critique is a way of approaching architecture's being. A developed architectural critique requires a theoretical background as a point of departure to look at and judge architecture. That's what I mean; architectural critique is of fundamental importance to a jury's evaluation of a competition proposal (M. Mäkinen, personal communication, 2006).

The jury must maintain a critical perspective and not begin to suggest improvements. The jury's assessment of competition proposals and appointment of a winner is a form of architectural critique. Attoe (1978) separates three different categories of criticism: *normative*, *interpretative* and *descriptive* criticism. Normative criticism deals with comparison based on a benchmark, which in the competition consists of demands, aims, and criteria in the competition programme. Interpretative criticism stems from experience. Descriptive criticism is oriented around facts and descriptions. All types of criticism can be found in the competition process in the Nordic Study. But we also found that professional jury members used architectural criticism as a *work method* for identifying and judging quality in competition entries (Rönn, 2010). As the jury examines the proposals more closely, critique takes on an educational function where merits, uncertainties, and shortcomings become apparent. Critique gives the jury members a better understanding of the design problems.

Discussion, findings and conclusion

In this article, I have tried to show what architectural quality means in a Nordic context – the way design criteria are used in a dialogue-based assessment following a special procedure typical of architectural competitions in the period, 1999-2000. This is fundamental to understanding how jury members representing different parties can select a single winner, which is a value embedded task, and still attain a consensus in their statement. The aim has been to illustrate and provide deeper knowledge about how qualified jurors go about judging the quality of an architectural project. The task is to identify a winner. Usually the jury succeeds.

I have closely read and analysed interview material in order to understand how architects regard quality in architectural competitions. The findings can be summarized as follows:

1. Architectural quality is an overall key concept. The concept in competitions is dominated by the aesthetic dimensions. Technical design is secondary. This can be explained by the fact that competitions take place at an early stage when the production of ideas is central to both the design and the evaluation of proposals. The concept is also value laden. This means that quality in architecture and urban design represents something positive and desirable. The concept appears to be professionally shaped and has been developed by education, sharpened by practice. Typical for architects also, is the idea that quality is linked to places and to how design proposals interact with their surroundings. There is a unique side to architecture. Architectural quality is attributed to proposals that address the site in a clear manner. The long life span of buildings explains why the locations play such a central role in quality assessment.

2. Design criteria have an open character and invite dialogue. The goal for the jury is to find the best solution for the competition task. The assessment process is driven forward by the jury addressing questions about the proposals. The jury communicates with drawings and illustrations. They seem to provide answers. It becomes a two-way communication between assessors and architectural representations. The assessment is a learning process. The questions concerning the design generate knowledge about the solutions. Weaknesses in the proposals and uncertainties are revealed, often as details. Dimensions of quality, on the other hand, are perceived as a whole. It is very seldom that one entry has a solution to the design problem in the brief, which is superior to all other proposals. Usually there are several good solutions to every design problem. This is why the design criteria have an open character. This helps the jury to understand the nature of the task by addressing questions about the proposals. It is the expression of reflection in action – a dialogue-based evaluation.

3. The assessment process is organized in a number of phases with specific activities. It takes about five jury meetings to come up with a winner in the Nordic countries. The favourites will stay in the competition until the final decision. The jury begins the assessment by a formal control of the proposals: Are they submitted on time? Do they provide enough information and meet the requirements? More important for the architectural quality are the following steps. After a first selection, the architects in the jury are usually asked to present the proposals to the jury. Then the jury gradually reduces the number of proposals. It is not difficult to single out a handful of good proposals with interesting solutions to the task. At this point, however, the assessment changes character. The good proposals become tools for learning about the competition task and its design problems. The favourite proposals shed light on the competition's brief. The winner is found in the final assessment when the task has been clarified by using architectural critique as a working method for identifying quality features and shortcomings. If it is still difficult to find a winner the jury has scale models made and tests the entries on the plot. The jury members can see with their own eyes that one of the remaining proposals fits the site better than the others and hence, can agree on a first prize winner.

I have presented principles of how quality issues are evaluated in competitions and the way the process works in the Nordic countries. These principles form a point of departure for a theory on judging entries. The assessment process has been broken down into typical stages. Even if architects are in a minority position on the jury, they have a great impact on assessment. They are supposed to be experts in the field of architecture and urban design. And as a

professional minority, the laypersons in the jury expect architects to guide them and point out the strengths and weaknesses in the designs. I have found that if a client's representative on the jury has had any doubts about architects, they are normally allayed by competence and consensus. Power, in this context, is not a question of numbers of members in the jury. The conclusion is that the organizing body's representatives rely on the good judgment made by recognized and skilled architects in the jury. Consensus is an indication that the jury has accomplished its mission and nominated a worthy winner of the competition. Both the design criteria and the model for the quality assessment seem to support consensus among jury members representing parties with different interests in the competition.

Magnus Rönn
Associate Professor, PhD
School of Architecture and Built Environment,
Royal Institute of Technology, Stockholm, Sweden
E-mail: magnus.ronn@arch.kth.se

References

Arge, K., & Bleiklie, S. (2003). *Arkitektonisk kvalitet - En studie om samspillet mellom byggherre og arkitekt*, Oslo: Norsk Form.

Attoe, W. (1978). *Architecture and critical imagination*. Chichester: John Wiley & Sons.

Collins, P. (1971). *Architectural Judgement.* London: Faber & Faber.

Drake, J. (1979*).* The Primary Generator and the Design Process. *Design Studies, 1*(1), 36-44.

European Forum for Architectural Policies (2011). *European Forum for Architectural Policies.* Retrieved June 14, 2011, from www.efap-fepa.eu/indexb.php?lg=en

European Parliament and of the Council (2004). *Directive 2004/18/EC on the coordination of procedures for the award of public works contracts, public supply contracts and public service contracts,* Article 73. Retrieved June 14, 2011, from http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:32004L0018:EN:HTML

Kaipiainen, M. (2006). *130 years of Finnish architectural competitions,* Helsinki: Finnish Association of Architects.

Kazemian, R., Rönn, M., & Svensson, C. (2005). *Jämförande analys av arkitekttävlingar – erfarenheter från tre nordiska länder*, Stockholm: TRITA-ARK-Forskningspublikationer 2005:3, KTH.

Kazemian, R., Rönn, M., & Svensson, C. (2007). *Arkitekturtävlingar i Finland – Erfarenheter från Finland,* Stockholm: Axl Books.

Kreiner, K. (2009). Architectural Competitions – Empirical Observations and Strategic Implications for Architectural Firms. *Nordic Journal of Architectural Research, 22*(2/3), 37-51.

Kuhn, T. (1970). *The Structure of Scientific Revolutions*. 2nd ed. Chicago: University of Chicago Press.

Kulturdepartementet (1995). *Kulturpolitikens inriktning*. SOU 1995:84. Stockholm: Fritzes.

Kulturdepartementet (1997). *Framtidsformer (Forms for the future): Förslag till handlingsprogram för arkitektur & formgivning*. SOU 1997:86. Stockholm: Fritzes. Retrieved June 14, 2011, from http://www.ud.se/sb/d/108/a/399

Lipstadt, H. (Ed.) (1989). *The Experimental Tradition*. New York: Princeton Architectural Press.

Nasar, J. L., (1999). *Design by Competitions*. Cambridge: Cambridge University Press.

Nashed, F. (2005). *Architectural Quality Control. An Illustrated Guide*. New York: McGraw-Hill.

Nelson, C. (2006). *Managing Quality in Architecture. A Handbook for Creators of the Built Environment.* Oxford: Architectural Press. Elsevier.

Östman, L. (2005). *A Pragmatist Theory of Design*, Stockholm: TRITA-ARK-Akademisk avhandling 2005:1, KTH.

Rittel, H., & Webber, M., (1984). Dilemmas in a General Theory of Planning. In Cross N., (Ed.), *Developments in Design Methodology* (pp. 155-169). New York: John Wiley & Sons.

Polanyi, M. (1983 [1966]). *The Tacit Dimension*. Glouchester, MA: Peter Smith.

Rönn, M. (2007). Kvalitetsföreställningar i arkitektur. I Rönn, M. (Red) *En fråga om kvalitet* (ss. 23-59) Estland: Santerus Förlag.

Rönn, M., (2010). Kritik, bedömning och beslut i arkitektur och stadsbyggnad - hur vinnare utses i arkitekturtävlingar. I *GPMS Proceedings,* GRI-rapport 2010:2. Göteborg: Göteborgs Universitet.

Rönn, M., Kazemian, R., & Andersson, J. E., (2010). (Eds.) *The architectural Competition. Research Inquiries and Experiences.* Latvia: Axl Books.

Socialdepartementet (1987). *Plan- och bygglag*. SFS 1987:10. Stockholm: SD. Retrieved June 14, 2011, from http://www.notisum.se/rnp/sls/lag/19870010.HTM

Svensson, C., (2009). Speaking of architecture. A study of the jury's assessment in an invited competition. *Nordic Journal of Architectural Research, 22*(2/3), 94-107.

Tostrup, E. (1999). *Architecture and rhetoric: text and design in architectural competitions, Oslo 1939-1997.* London: Andreas Papadakis.

[1] Architecture policy programs in Europe: Netherlands; Space for Architecture (1991), Architecture of Space (1996), Constructing the Netherlands (2001) and Action Programme Space and Culture (2005). Norway; Surroundings as Culture, (1992), Aesthetics in Government Buildings and Constructions, (1997) and Architecture Now, (2009). Denmark; Danish Architecture, (1994), Architecture 1996 (1996) and Nation of Architecture Denmark, (2007). Sweden; Forms for the future (1997), Finland; Finland's Architectural Policy (1998), Germany; Building Culture in Germany (2001). Ireland; Action on Architecture (2002). Austria; the Austrian Report on Building Culture (2006). Scotland, Building Our Legacy, Statement on Scotland's architecture policy (2007).

[2] Thomas Kuhn (1970), who refers to Michael Polanyi's (1983 [1966]) concept tacit knowing.

[3] Finland has a different tradition from the other Nordic countries for invited competition. In Finland it is the competitors that choose the architect judges.

[4] According to § 10 of Finnish competitions regulation 2007 a proposal can be disqualified for (a) not keeping confidentiality, (b) not being delivered in time and as is requested by the competition programme (c) being unanimously considered immature by the jury, (d) failing to comply with actions which the programme demands.